



COMPUTER ARCHITECTURE LAB



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Semester: 4th Semester

Task # 1:

Circuit Design of 4:1 Multiplexer:

Truth Table:

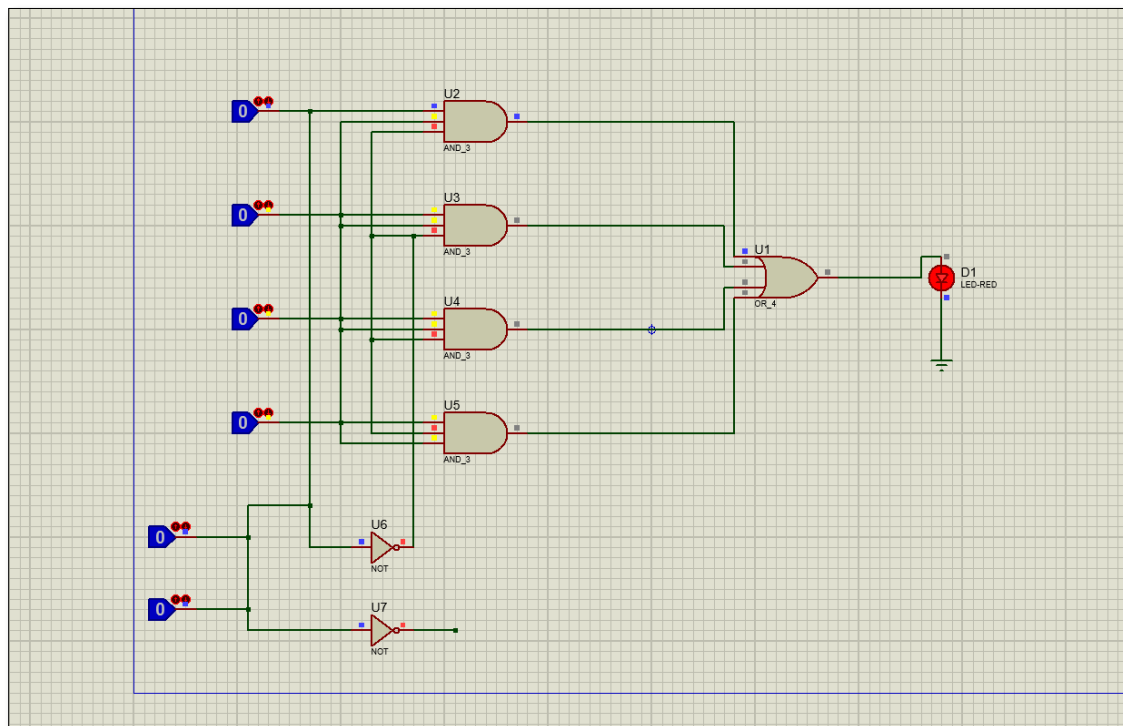
Selection Lines		Output
S0	S1	Y
0	0	D ₀
0	1	D ₁
1	0	D ₂
1	1	D ₃

Boolean Expression:

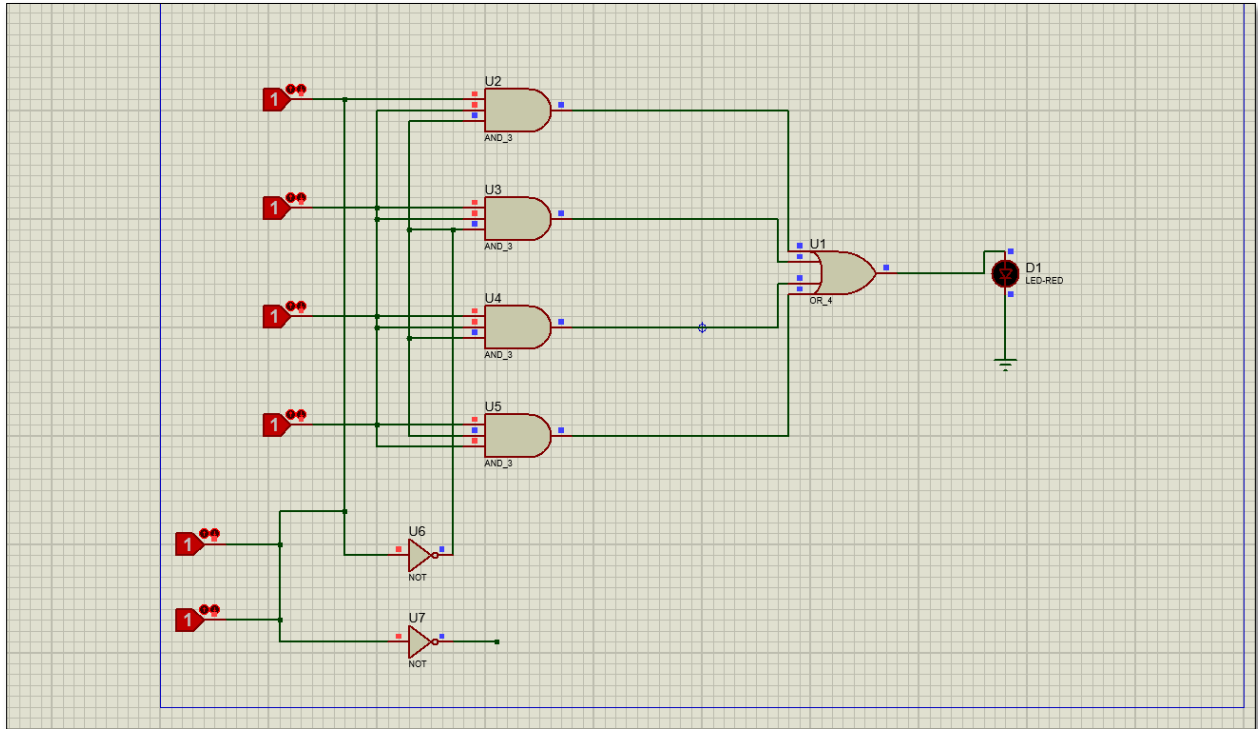
$$Y = (I_1 * S_1' * S_0') + (I_2 * S_1' * S_0) + (I_3 * S_1 * S_0') + (I_4 * S_1 * S_0)$$

Circuit Diagram:

On State:



OFF State:



Task # 2:

Circuit Design of 2:1 Multiplexer:

Truth Table:

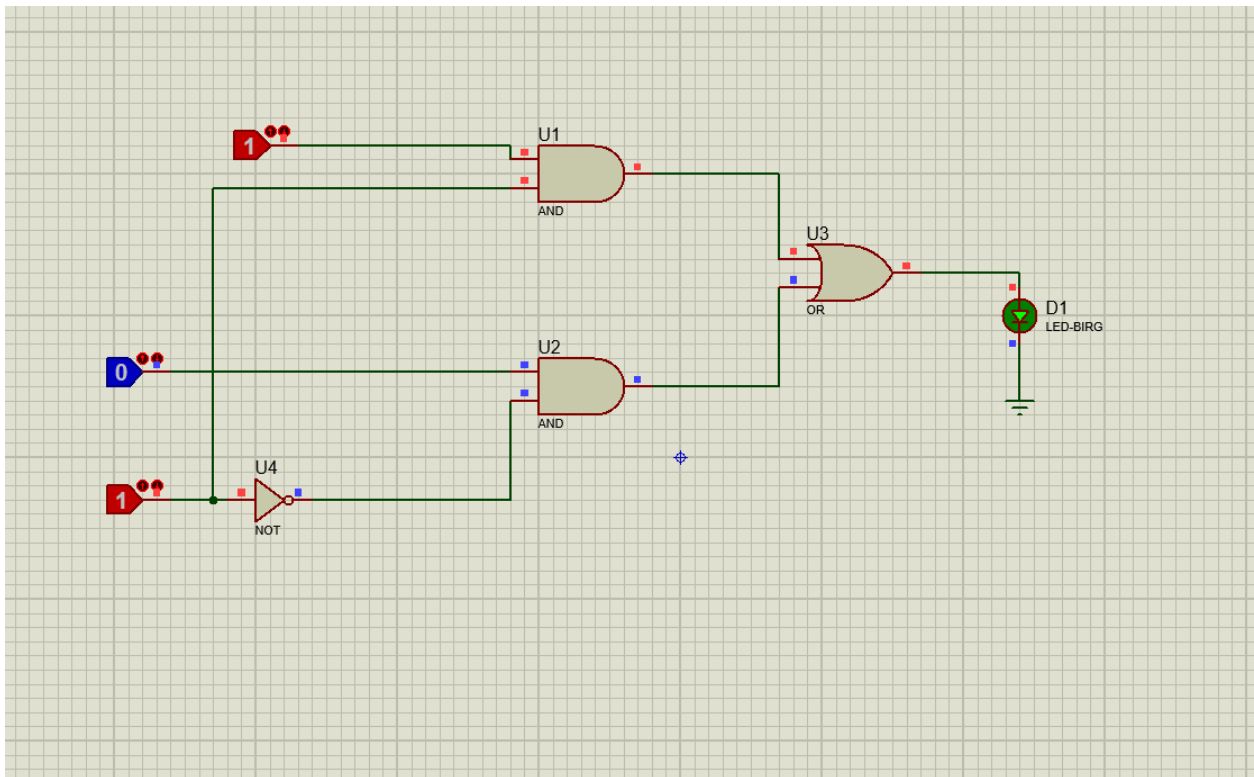
S_0	I_0	I_1	Y
0	0	X	0
0	1	X	1
1	X	0	0
1	X	1	1

Boolean Expression:

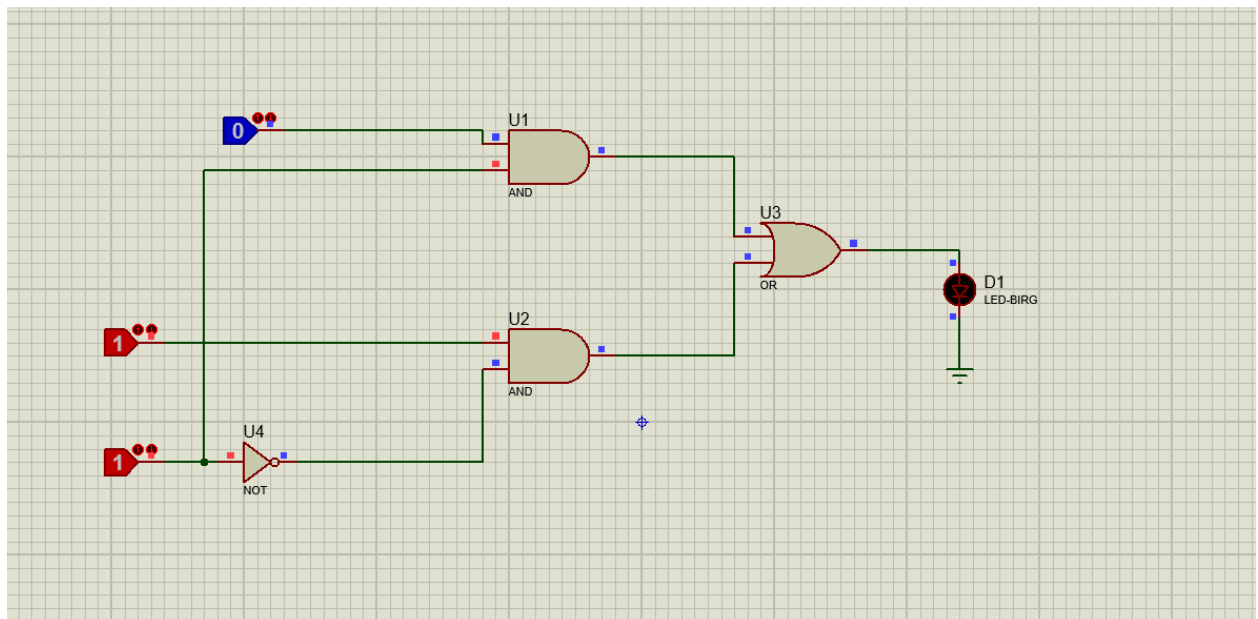
$$Y = I_0 \bar{S} + I_1 S$$

Circuit Diagram:

On State:



OFF State:



Task # 3:

Circuit Design of 8:1 Multiplexer:

Truth Table:

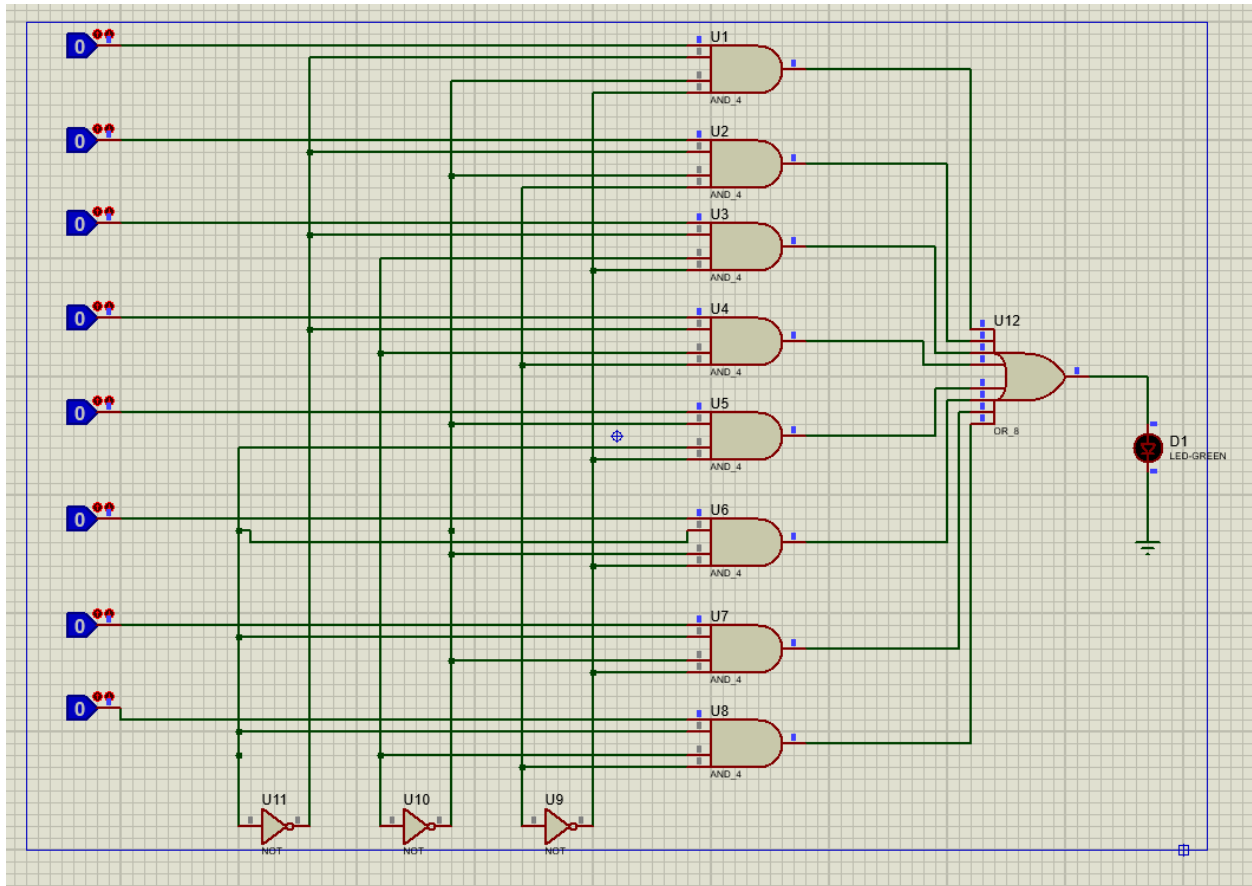
INPUTS			Output
S_2	S_1	S_0	Y
0	0	0	A_0
0	0	1	A_1
0	1	0	A_2
0	1	1	A_3
1	0	0	A_4
1	0	1	A_5
1	1	0	A_6
1	1	1	A_7

Boolean Expression:

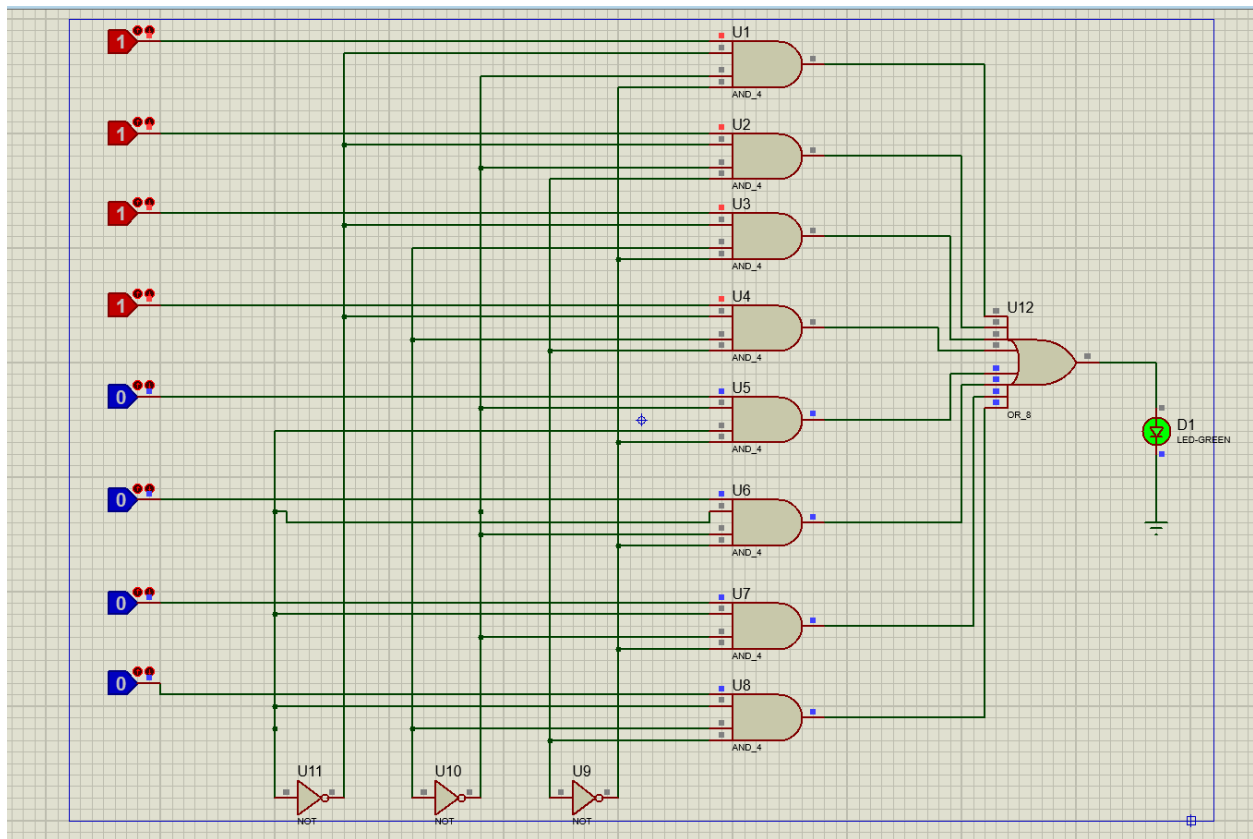
$$Y = S_0' \cdot S_1' \cdot S_2' \cdot A_0 + S_0 \cdot S_1' \cdot S_2' \cdot A_1 + S_0' \cdot S_1 \cdot S_2' \cdot A_2 + S_0 \cdot S_1 \cdot S_2' \cdot A_3 + S_0' \cdot S_1' \cdot S_2 \cdot A_4 + S_0 \cdot S_1' \cdot S_2 \cdot A_5 + S_0' \cdot S_1 \cdot S_2 \cdot A_6 + S_0 \cdot S_1 \cdot S_2 \cdot A_7$$

Circuit Diagram:

OFF State:



ON State:



Task # 4:

Circuit Design of 1:4 DeMultiplexer:

Truth Table:

Input	Select Lines	Output Lines
I	$S_1 S_0$	$D_0 D_1 D_2 D_3$
I	0 0	1 0 0 0
I	0 1	0 1 0 0
I	1 0	0 0 1 0
I	1 1	0 0 0 1

Boolean Expression:

$$I_0 = S_1' S_0' D$$

$$I_1 = S_1' S_0 D$$

$$I_2 = S_1 S_0' D$$

$$I_3 = S_1 S_0 D$$

Circuit Diagram:

OFF State:

